THE INTEGER 1117175146 IS NOT THE SUM OF A PRIME AND AT MOST THREE POWERS OF TWO

MATH0

Abstract. We prove that $N = 1117175146$ cannot be written as $p+2^{e_1}+\cdots+2^{e_k}$ with p prime, $0 \le k \le 3$, and exponents $e_1, \ldots, e_k \ge 0$ that need not be distinct. This was observed by Bogdan Grechuk in connection with Erdős Problem #10, and shows that the restriction to odd integers in a conjecture of Granville and Soundararajan cannot simply be dropped; the result is known, and we provide a complete certificate-based proof and, to our knowledge, its first formal proof. The proof bounds every exponent by 30 and checks, for each of the 5828 admissible exponent multisets with sum s, that $N - s$ has a proper divisor in an explicit list of 38 primes; we also give a hand-checkable form of this certificate based on $N - 1 = (2^{16} - 1) \cdot 17047$. The argument is verified in Lean 4, with the finite computation carried out by the Lean kernel, against the statement `Erdos10.erdos_10.variants.grechuk_example` of the formal-conjectures repository.

1. Introduction

Erdős asked whether there is a constant k such that every sufficiently large integer is the sum of a prime and at most k powers of 2 [4, p. 96], [5, p. 28]. This is Problem #10 on the Erdős Problems website [1], where it is listed as open. Gallagher [7] proved that for every $\varepsilon > 0$ there is $k(\varepsilon)$ such that the integers that are the sum of a prime and at most $k(\varepsilon)$ powers of 2 have lower density at least $1 - \varepsilon$. Granville and Soundararajan [8] conjectured that at most three powers of 2 suffice for all odd integers, and hence at most four suffice for all even integers.

As recorded on [1], the restriction to odd integers matters: Bogdan Grechuk observed that 1117175146 is not the sum of a prime and at most three powers of 2. Grechuk also pointed out that parity considerations, together with the existence of many integers that are not the sum of a prime and two powers of 2, suggest that infinitely many even integers are not the sum of a prime and at most three powers of 2. The formal-conjectures repository [6] notes that this infinite statement follows from Crocker's construction [2] of infinitely many odd integers that are not the sum of a prime and two powers of 2, and links a Lean formalisation of the infinite statement [3]; that formalisation does not treat the specific integer 1117175146.

Status and contribution. The statement proved here is Grechuk's observation and is not new. The page [1] records it without describing how it was verified. We give (i) a complete proof by an explicit finite certificate (Section 3); (ii) a form of the certificate that can be checked by hand, in which all but 44 of the remainders are handled by one congruence argument (Section 3.3); and (iii) a Lean 4 proof of the upstream formal statement `Erdos10.erdos_10.variants.grechuk_example` of [6] (Section 4). The upstream file links no formal proof for this statement. The Lean files it links as formal proofs of two neighbouring variants (the infinite statement above [3], and a statement about two powers of 2 [11]) do not prove it: the first does not mention 1117175146, and the second restates it with `sorry`. A web search found no other formal proof. We therefore classify the result as the *first formal proof of a known result*. We do not know how Grechuk found or verified the example.

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2. Statement

Let $\mathbb{N} = \{0, 1, 2, \dots\}$. For $k \in \mathbb{N}$ let

$$\mathcal{S}_k = \Big\{ p + \sum_{i=1}^{j} 2^{e_i} \ : \ p \text{ prime}, \ 0 \le j \le k, \ e_1, \dots, e_j \in \mathbb{N} \Big\}.$$

Here $2^0 = 1$ counts as a power of two, and the exponents need not be distinct.

Theorem 1. *Let $N = 1117175146$. Then $N \notin \mathcal{S}_3$; that is, there are no prime p, integer $0 \le j \le 3$ and exponents $e_1, \dots, e_j \in \mathbb{N}$ with $N = p + 2^{e_1} + \cdots + 2^{e_j}$.*

Because repeated exponents, the exponent 0, and fewer than three summands are all allowed, Theorem 1 implies the variants in which the powers are required to be distinct, to be at least 2, or to be exactly three in number.

In the formal-conjectures repository [6] (commit `c77fdd4`) the set $\mathcal{S}_k$ is the abbreviation `sumPrimeAndTwoPows` k, in which the exponents form a multiset `pows` of natural numbers of cardinality at most k. A multiset of cardinality j is the same as an unordered j-tuple, so this is exactly the set $\mathcal{S}_k$ above. The definition and the statement, copied verbatim from `FormalConjectures/ErdosProblems/10.lean`, are:

```
/--
The set of natural numbers that can be written as a sum
of a prime and at most $k$ powers of $2$.
-/
abbrev sumPrimeAndTwoPows (k : ℕ) : Set ℕ :=
  { p + (pows.map (2 ^ ·)).sum | (p : ℕ) (pows : Multiset ℕ) (_ : p.Prime)
    (_ : pows.card ≤ k)}

/--
Bogdan Grechuk has observed that `1117175146` is not the sum of a prime
and at most $3$ powers of $2$.
-/
@[category research solved, AMS 5 11]
theorem erdos_10.variants.grechuk_example :
    1117175146 ∉ sumPrimeAndTwoPows 3 := by
  sorry
```

3. Proof

Throughout, $N = 1117175146$. Its binary expansion is

$$N = 2^{30} + 2^{25} + 2^{23} + 2^{20} + 2^{18} + 2^{17} + 2^{15} + 2^{13} + 2^{12} + 2^{11} + 2^{10} + 2^8 + 2^6 + 2^5 + 2^3 + 2^1,$$

so in particular $2^{30} \le N < 2^{31} = 2147483648$.

3.1. The certificate.

Let W be the following list of 38 primes:

$$\begin{aligned} W = (&2, 3, 5, 7, 11, 13, 17, 19, 23, 37, 41, 47, 59, 61, 73, 79, 89, 97, 101, \\ &103, 107, 131, 163, 199, 211, 229, 257, 379, 547, 659, 877, 1381, 1481, \\ &1699, 2099, 7283, 7703, 19391). \end{aligned}$$

Definition 2. An integer $m \in \mathbb{N}$ is *certified* if $m < 2$, or if there is $q \in W$ with $q \mid m$ and $q < m$.

Lemma 3. *A certified integer is not prime.*

Proof. If $m < 2$ then m is not prime. Otherwise some $q \in W$ satisfies $q \mid m$ and $q < m$. Every element of W is at least 2, so q is a divisor of m with $1 < q < m$, and m is not prime. □

The proof uses only that every element of W is at least 2; the primality of the elements of W plays no role, and the formal proof does not check it.

Lemma 4. *If $e \in \mathbb{N}$ and $2^e \le N$, then $e \le 30$.*

Proof. If $e \ge 31$ then $2^e \ge 2^{31} > N$. □

Proposition 5. *Let $j \in \{0,1,2,3\}$ and $e_1, \dots, e_j \in \mathbb{N}$, and put $s = \sum_{i=1}^{j} 2^{e_i}$. If $s \le N$, then $N - s$ is certified.*

Proof. Each summand satisfies $1 \le 2^{e_i} \le s \le N$, so $e_i \le 30$ by Lemma 4. It therefore suffices to check the claim for the tuples $(e_1, \dots, e_j) \in \{0, \dots, 30\}^j$ with $s \le N$: the empty tuple ($s = 0$), the 31 singletons, 952 of the $31^2 = 961$ ordered pairs, and 29009 of the $31^3 = 29791$ ordered triples. For each of these one verifies by trial division that $N - s$ has a divisor $q \in W$ with $q < N - s$. This finite verification is carried out by the Lean kernel (Section 4); Section 3.3 gives a version of it that can be checked by hand. □

Up to reordering there are 5984 multisets of at most three exponents in $\{0, \dots, 30\}$, and 5828 of them satisfy $s \le N$. Of these, 5336 have s even, and then $q = 2$ is a witness (see Section 3.3). The other 492 have s odd; for them the least prime factor of $N - s$ is always a witness, and W consists exactly of 2 together with the 37 distinct least prime factors that occur. The least prime factor is 3 in 224 of these cases, 5 in 111, 17 in 40 and 7 in 28, while the largest primes in W, namely 7283, 7703 and 19391, are each needed for only one or two exponent multisets.

3.2. **Conclusion.**

Proof of Theorem 1. Suppose, for a contradiction, that $N = p + \sum_{i=1}^{j} 2^{e_i}$ with p prime and $0 \le j \le 3$. (In the formal proof the multiset of exponents is represented by a list $(e_1, \dots, e_j)$, and the cases $j = 0, 1, 2, 3$ are treated separately; a list of length at least 4 contradicts the cardinality bound.) Put $s = \sum_{i=1}^{j} 2^{e_i}$. Since $p \ge 0$ we have $s \le N$ and $p = N - s$. By Proposition 5, p is certified, so p is not prime by Lemma 3. This contradiction proves Theorem 1. □

3.3. **A form of the certificate that can be checked by hand.**

This subsection is not used in the formal proof. It gives a second derivation of Proposition 5, organised so that every step can be checked by hand or with a pocket calculator. For $n \in \mathbb{N}$ let $\nu(n)$ be the number of ones in the binary expansion of n.

Lemma 6. *If $s = 2^{e_1} + \cdots + 2^{e_j}$, then $\nu(s) \le j$.*

Proof. It suffices to show $\nu(n + 2^e) \le \nu(n) + 1$ for all $n, e \in \mathbb{N}$ and then induct on j. If bit e of n is 0, adding 2^e sets it, and ν increases by 1. Otherwise let $t \ge 1$ be the length of the maximal run of ones in n starting at bit e; adding 2^e clears these t bits and sets bit $e + t$, so ν changes by $1 - t \le 0$. □

A direct computation gives

$$\nu(N) = 16, \quad \nu(N-1) = 16, \quad \nu(N-2) = 15, \quad \nu(N-3) = 17,$$
$$\nu(N-5) = 16, \quad \nu(N-17) = 16, \quad \nu(N-257) = 15.$$

By Lemma 6, a sum s of at most three powers of 2 is therefore never equal to $N - r$ for $r \in \{0, 1, 2, 3, 5, 17, 257\}$; that is,

$$N - s \notin \{0, 1, 2, 3, 5, 17, 257\}. \tag{1}$$

In particular $N - s \ge 2$, so the alternative $m < 2$ in Definition 2 never occurs.

Even s. Then $N - s$ is even, and by (1) it is neither 0 nor 2. So $2 \mid N - s$ and $2 < N - s$.

Odd s. Since $2^{e_i} \le s \le N$, every exponent e_i is at most 30 by Lemma 4. The parity of s is the parity of the number of indices i with $e_i = 0$, so this number is 1 or 3. If it is 3, then $s = 3 = 1 + 2^1$. If it is 1, then $s = 1 + t$, where t is 0, a single power 2^c with $1 \le c \le 30$, or a sum $2^a + 2^b$ with $1 \le a \le b \le 30$. If $a = b$ then $t = 2^{a+1}$ with $a+1 \le 30$, because $2^{a+1} \le s \le N$. If $a < b$ and $16 \mid b-a$, then $b = a+16$, because $b - a \le 29$. Hence $N - s$ is one of the following:

(a) $N - 1$;
(b) $N - 1 - 2^c$ with $1 \le c \le 30$;
(c) $N - 1 - 2^a - 2^b$ with $1 \le a < b \le 30$ and $b \not\equiv a \pmod{16}$;
(d) $N - 1 - 2^a - 2^{a+16}$ with $1 \le a \le 14$.

The key fact is the factorisation

$$N - 1 = 1117175145 = (2^{16} - 1) \cdot 17047, \qquad 2^{16} - 1 = 3 \cdot 5 \cdot 17 \cdot 257.$$

Lemma 7. *In cases* (a) *and* (c), *$N - s$ is divisible by some $q \in \{3, 5, 17, 257\}$ with $q < N - s$.*

Proof. In case (a), $N - s = N - 1$, and $3 \mid 2^{16} - 1 \mid N - 1$ with $3 < N - 1$. In case (c), write $b - a = 2^v u$ with u odd; since $16 \nmid b - a$ we have $v \in \{0, 1, 2, 3\}$. Let $q = 2^{2^v} + 1 \in \{3, 5, 17, 257\}$, a divisor of $2^{16} - 1 = (2 + 1)(2^2 + 1)(2^4 + 1)(2^8 + 1)$. Then $2^{2^v} \equiv -1 \pmod{q}$, hence $2^{b-a} \equiv (-1)^u = -1 \pmod{q}$, and so q divides $2^a(2^{b-a} + 1) = 2^a + 2^b$. As $q \mid N - 1$, we get $q \mid N - 1 - 2^a - 2^b = N - s$. Finally $N - s \ne q$ by (1), so $q < N - s$. □

Case (c) accounts for 417 of the 492 odd-s multisets and case (a) for one more. The remaining 74 multisets lead to only 44 distinct remainders, those of cases (b) and (d). Table 1 lists each of them with a prime divisor q (its least prime factor) and the cofactor; in every row $1 < q < N - s$. All primes used in Lemma 7 and in Table 1 belong to W. This proves Proposition 5 again.

c	$N - 1 - 2^c$	q	cofactor	c	$N - 1 - 2^c$	q	cofactor
1	1117175143	7	159596449	16	1117109609	7	159587087
2	1117175141	1381	808961	17	1117044073	89	12551057
3	1117175137	13	85936549	18	1116913001	163	6852227
4	1117175129	7	159596447	19	1116650857	7	159521551
5	1117175113	23	48572831	20	1116126569	19391	57559
6	1117175081	11	101561371	21	1115077993	59	18899627
7	1117175017	7	159596431	22	1112980841	7	158997263
8	1117174889	379	2947691	23	1108786537	107	10362491
9	1117174633	37	30193909	24	1100397929	101	10895029
10	1117174121	7	159596303	25	1083620713	7	154802959
11	1117173097	1481	754337	26	1050066281	11	95460571
12	1117171049	73	15303713	27	982957417	13	75612109
13	1117166953	7	159595279	28	848739689	7	121248527
14	1117158761	1699	657539	29	580304233	2099	276467
15	1117142377	13	85934029	30	43433321	73	594977
a	$N - 1 - 2^a - 2^{a+16}$	q	cofactor	a	$N - 1 - 2^a - 2^{a+16}$	q	cofactor
1	1117044071	11	101549461	8	1100397673	19	57915667
2	1116912997	877	1273561	9	1083620201	41	26429761
3	1116650849	101	11055949	10	1050065257	7703	136319
4	1116126553	659	1693667	11	982955369	11	89359579
5	1115077961	47	23725063	12	848735593	547	1551619
6	1112980777	7283	152819	13	580296041	13	44638157
7	1108786409	79	14035271	14	43416937	131	331427

TABLE 1. The 44 remainders of cases (b) and (d), with least prime factor q and cofactor.

```
def good (m : ℕ) : Bool := m < 2 || wit.any (fun q => m % q == 0 && q < m)

def check3 : Bool :=
  (List.range 31).all fun x => (List.range 31).all fun y => (List.range 31).all fun z =>
    !(2 ^ x + 2 ^ y + 2 ^ z ≤ 1117175146) || good (1117175146 - (2 ^ x + 2 ^ y + 2 ^ z))

theorem check3_eq : check3 = true := by decide +kernel

theorem erdos_10_grechuk_example : type_of% @Erdos10.erdos_10.variants.grechuk_example := by
  rintro ⟨p, pows, hp, hcard, h⟩
  induction pows using Quotient.inductionOn with
  | h l =>
  rcases l with _ | ⟨a, _ | ⟨b, _ | ⟨c, _ | ⟨d, l⟩⟩⟩⟩
  · simp at h
    exact not_prime_of_good (h ▷ good_N) hp
  · simp at h
    have hs : 2 ^ a ≤ 1117175146 := by omega
    exact not_prime_of_good (show p = 1117175146 - 2 ^ a by omega ▷ spec1 hs) hp
  · simp at h
    have hs : 2 ^ a + 2 ^ b ≤ 1117175146 := by omega
    exact not_prime_of_good (show p = 1117175146 - (2 ^ a + 2 ^ b) by omega ▷ spec2 hs) hp
  · simp at h
    have hs : 2 ^ a + 2 ^ b + 2 ^ c ≤ 1117175146 := by omega
    exact not_prime_of_good (show p = 1117175146 - (2 ^ a + 2 ^ b + 2 ^ c) by omega ▷ spec3 hs)
    hp
  · simp at hcard
    omega

#math0_proves Math0.Erdos10.erdos_10_grechuk_example Erdos10.erdos_10.variants.grechuk_example
```

FIGURE 1. Verbatim excerpt of the Lean file.

In particular, the largest elements of W come from Table 1: 19391 is needed for $s = 1+2^{20} = 1+2^{19}+2^{19}$, 7283 for $s = 1+2^6+2^{22}$, and 7703 for $s = 1+2^{10}+2^{26}$.

4. Machine verification

Files and declarations. The proof is in the following file of the math0 repository, `https://github.com/Foundation-0/math0`:

`lean/Math0/ErdosProblems/Erdos10_erdos_10_variants_grechuk_example.lean`,

in the namespace `Math0.Erdos10`. The main theorem is `Math0.Erdos10.erdos_10_grechuk_example`. It is stated as `type_of% @Erdos10.erdos_10.variants.grechuk_example`, so its type is, by construction, the type of the upstream declaration. The auxiliary declarations are `wit` (W), `good` (Definition 2, as a Boolean function), `not_prime_of_good` (Lemma 3), `exp_lt` (Lemma 4), the Boolean checks `check1`–`check3` over $\{0,\dots,30\}^j$ with their kernel evaluations `check1_eq`–`check3_eq` and `good_N` ($j=0$), and `spec1`–`spec3`, which combine these with `exp_lt` into Proposition 5. Figure 1 reproduces the certificate test, the main proof and the final audit command verbatim. The main proof follows Section 3: the multiset of exponents is replaced by a list representing it (via `Quotient.inductionOn`); the list is split by length, `omega` derives $s \le N$ and $p = N - s$ from $p + s = N$, and the certificate gives $\neg\,\mathrm{Prime}(p)$. In the last case `simp` rewrites the cardinality hypothesis as an inequality between numerals and the list length that `omega` refutes.

Kernel computation. The finite verification consists of the four theorems `good_N`, `check1_eq`, `check2_eq` and `check3_eq`, each proved by `decide +kernel`. This tactic closes a decidable goal with a proof term whose type-correctness the Lean kernel establishes by reducing the decision procedure to `true`, evaluating the Boolean functions by definitional unfolding with its built-in arbitrary-precision arithmetic on natural-number literals. No compiled code is trusted: the file does not use `native_decide`, which would introduce the axiom `Lean.ofReduceBool`. The largest check, `check3_eq`, runs through all 29791 triples and takes about 20 seconds; the whole file compiles in about 30 seconds with default settings.

Audit. The last command of the file is `#math0_proves`, applied to our theorem and the upstream declaration (Figure 1). This command, defined in `lean/Math0/Audit.lean`, fails unless (1) the two declarations have the same universe parameters, (2) their types are syntactically identical after removing metadata annotations, with no unfolding and no definitional-equality check, and (3) our theorem depends on no axioms other than `propext`, `Classical.choice` and `Quot.sound`. In particular it rejects `sorryAx`, `Lean.ofReduceBool` and `Lean.trustCompiler`, and any custom axiom. On this file it reports

```
math0 audit: 'Math0.Erdos10.erdos_10_grechuk_example' depends on axioms [Classical.choice, Quot.
    sound, propext]
math0 statement check: 'Math0.Erdos10.erdos_10_grechuk_example' proves 'Erdos10.erdos_10.
    variants.grechuk_example'
```

Toolchain and reproduction. We use Lean 4 [10] `v4.33.1`, Mathlib [9] `v4.33.1` and formal-conjectures [6] at commit `c77fdd4`, all pinned in `lean/`. To reproduce, run `scripts/verify.sh` from the repository root. It fetches the Mathlib build cache, builds all math0 results with `lake build` (re-running every audit command), checks that the `formal_conjectures` and `mathlib` checkouts are unmodified, and reports the number of passed statement checks. The single file can also be checked by running `lake env lean` on it from the `lean` directory. A recomputation in Python (SymPy) confirmed all counts above and that no remainder $N - s$ is prime.

References

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[11] anatoliiohorodnyk, *lean-fc-proofs*, GitHub repository, file `Proofs/T_Erdos10_erdos_10_variants_two_pows.lean` at commit `5da9642`, `https://github.com/anatoliiohorodnyk/lean-fc-proofs`, linked from [6].